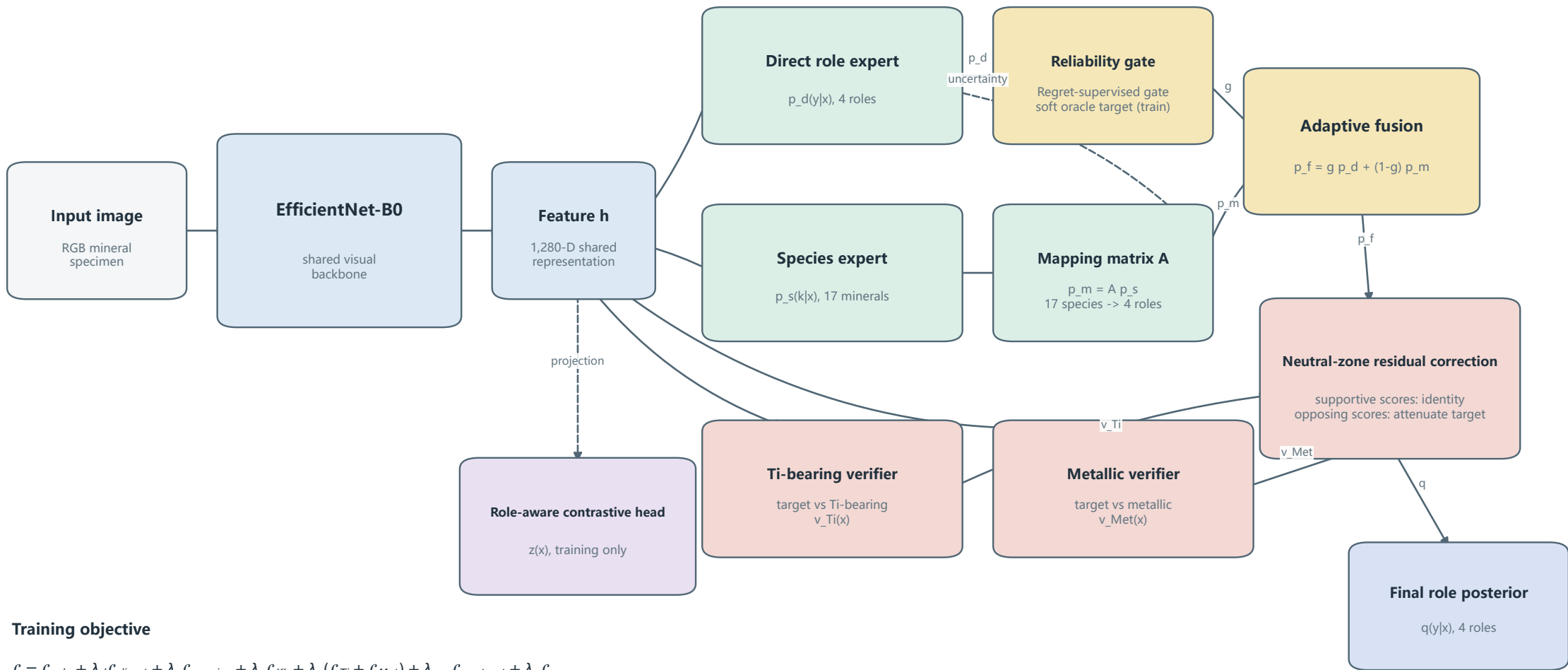


RSG-HRGV-Net: regret-supervised hierarchical evidence routing and risk verification

共享视觉表征支持角色/种类双专家；训练期软后悔目标监督门控选择更可靠的证据



Training objective

$$\mathcal{L} = \mathcal{L}_{role} + \lambda_d \mathcal{L}_{direct} + \lambda_s \mathcal{L}_{species} + \lambda_c \mathcal{L}_{KL} + \lambda_v (\mathcal{L}_{Ti} + \mathcal{L}_{Met}) + \lambda_{con} \mathcal{L}_{contrast} + \lambda_g \mathcal{L}_{reg}$$

Solid arrows: inference path Dashed arrows: uncertainty/training cues The regret target uses stop-gradient expert evidence; verifier coupling is retained in the formal model.

$$\Delta = \ell_m - \ell_d, \quad g^* = \sigma(\Delta/T_r), \quad w = \tanh(|\Delta|/T_w), \quad \mathcal{L}_{reg} = w \text{BCE}(g, g^*)$$

$$c_h(v_h) = [\tau_h - v_h]_+ / \tau_h, \quad q_T \propto p_f \tau \exp[-\beta_{Ti} C_{Ti} - \beta_{Met} C_{Met}], \quad q_j \propto p_{f,j}$$